PSEUDO-COHERENCE IS PRESERVED AND REFLECTED BY THE PUSHFORWARD ALONG A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The pushforward along a closed immersion of ringed spaces is exact, so the derived and the plain pushforward agree. used at proof lines 3151-3153. Lets the proof work with the plain pushforward throughout.
- A finitely generated cohomology module of a pseudo-coherent complex admits, locally, a surjection from a finite free module onto it. invoked at proof lines 3160-3164. Produces the map from a shifted free module onto the top cohomology group.
- In a distinguished triangle, if two terms are pseudo-coherent then so is the third, so cutting off the top cohomology preserves the property. invoked at proof lines 3166-3168 and again at proof lines 3170-3175. Gives that the cone is pseudo-coherent and then that the pushed forward complex is pseudo-coherent.

Source reference. Original lemma 01AX: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 3151-3153 as the fact used throughout that the derived pushforward is the plain one; all subsequent arguments are inside the counted span.

Source reference. Original lemma 08DN: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 3160-3162 to see that the top cohomology module is of finite type, which produces the surjection from a free module; the construction of the triangle that follows is inside the counted span.

Source reference. Original lemma 08CD: sources/stacks-chow/tags.txt, source lines 1–1. Applied twice, at proof lines 3166-3168 to see that the cone is pseudo-coherent and at proof lines 3172-3176 to transfer the property along the pushed forward triangle; both applications are inside the counted span.

Lemma 0.2. *Let $i : Z \to X$ be a morphism of ringed spaces such that i is a closed immersion of underlying topological spaces and such that $i_*\mathcal{O}_Z$ is pseudo-coherent as an $\mathcal{O}_X$-module. Let $E \in D(\mathcal{O}_Z)$. Then E is m-pseudo-coherent if and only if Ri_*E is m-pseudo-coherent.*

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Proof. Throughout this proof we will use that i_* is an exact functor, and hence that $Ri_* = i_*$, see Modules, Lemma 01AX.

Assume E is m-pseudo-coherent. Let $x \in X$. We will find a neighbourhood of x such that i_*E is m-pseudo-coherent on it. If $x \notin Z$ then this is clear. Thus we may assume $x \in Z$. We will use that $U \cap Z$ for $x \in U \subset X$ open form a fundamental system of neighbourhoods of x in Z. After shrinking X we may assume E is bounded above. We will argue by induction on the largest integer p such that $H^p(E)$ is nonzero. If $p < m$, then there is nothing to prove. If $p \geq m$, then $H^p(E)$ is an $\mathcal{O}_Z$-module of finite type, see Cohomology, Lemma 08DN. Thus we may choose, after shrinking X, a map $\mathcal{O}_Z^{\oplus n}[-p] \to E$ which induces a surjection $\mathcal{O}_Z^{\oplus n} \to H^p(E)$. Choose a distinguished triangle

$$\mathcal{O}_Z^{\oplus n}[-p] \to E \to C \to \mathcal{O}_Z^{\oplus n}[-p+1]$$

We see that $H^j(C) = 0$ for $j \geq p$ and that C is m-pseudo-coherent by Cohomology, Lemma 08CD. By induction we see that i_*C is m-pseudo-coherent on X. Since $i_*\mathcal{O}_Z$ is m-pseudo-coherent on X as well, we conclude from the distinguished triangle

$$i_*\mathcal{O}_Z^{\oplus n}[-p] \to i_*E \to i_*C \to i_*\mathcal{O}_Z^{\oplus n}[-p+1]$$

and Cohomology, Lemma 08CD that i_*E is m-pseudo-coherent.

Assume that i_*E is m-pseudo-coherent. Let $z \in Z$. We will find a neighbourhood of z such that E is m-pseudo-coherent on it. We will use that $U \cap Z$ for $z \in U \subset X$ open form a fundamental system of neighbourhoods of z in Z. After shrinking X we may assume i_*E and hence E is bounded above. We will argue by induction on the largest integer p such that $H^p(E)$ is nonzero. If $p < m$, then there is nothing to prove. If $p \geq m$, then $H^p(i_*E) = i_*H^p(E)$ is an $\mathcal{O}_X$-module of finite type, see Cohomology, Lemma 08DN. Choose a complex $\mathcal{E}^\bullet$ of $\mathcal{O}_Z$-modules representing E. We may choose, after shrinking X, a map $\alpha : \mathcal{O}_X^{\oplus n}[-p] \to i_*\mathcal{E}^\bullet$ which induces a surjection $\mathcal{O}_X^{\oplus n} \to i_*H^p(\mathcal{E}^\bullet)$. By adjunction we find a map $\alpha : \mathcal{O}_Z^{\oplus n}[-p] \to \mathcal{E}^\bullet$ which induces a surjection $\mathcal{O}_Z^{\oplus n} \to H^p(\mathcal{E}^\bullet)$. Choose a distinguished triangle

$$\mathcal{O}_Z^{\oplus n}[-p] \to E \to C \to \mathcal{O}_Z^{\oplus n}[-p+1]$$

We see that $H^j(C) = 0$ for $j \geq p$. From the distinguished triangle

$$i_*\mathcal{O}_Z^{\oplus n}[-p] \to i_*E \to i_*C \to i_*\mathcal{O}_Z^{\oplus n}[-p+1]$$

the fact that $i_*\mathcal{O}_Z$ is pseudo-coherent and Cohomology, Lemma 08CD we conclude that i_*C is m-pseudo-coherent. By induction we conclude that C is m-pseudo-coherent. By Cohomology, Lemma 08CD again we conclude that E is m-pseudo-coherent. □